SPE-229170-MS

Introduction of a Low-Carbon Geopolymer Alternative for Sustainable Cementing Applications across the Middle East: A UAE Case Study

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This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

Abstract

With the focus on sustainability in oilfield drilling operations, many initiatives are being implemented across the industry to lower carbon emissions and reduce the overall environmental footprint. One such initiative involves transitioning away from Portland cement in well construction and replacing it with a low-carbon alternative. Across the Middle East, an alternative geopolymer system has been successfully introduced, with case histories in Oman, the UAE, Saudi Arabia, and Qatar. This paper focuses on case histories from the UAE.

Portland cement has been used as a well barrier in the oilfield for over 100 years. However, its manufacturing process is energy-intensive and results in significant CO2 emissions. By introducing an alternative solution, the CO2footprint can be substantially reduced while the properties of Portland cement can be matched or even improved. To facilitate the implementation of the geopolymer system, the job design and execution must remain similar to current cementing practices. Additionally, the liquid and set properties of the geopolymer system must be comparable to, or better than, the properties of the Portland cement systems currently used for each operation.

The introduction of the low-carbon geopolymer system begins with identifying local sources of suitable aluminosilicates as the primary material. To minimize transport-related emissions, a local source was identified in each country. Extensive laboratory testing was performed to qualify the raw material and produce a fluid design that meets the requirements for wellbore conditions. Standard cementing lab equipment and procedures were used for laboratory testing. Additives were either dry-blended or added to the mix fluid, offering a wide variety of options to achieve the required fluid properties.

Local yard testing demonstrated that the geopolymer system could be batch-mixed or fly-mixed as required by rig operations and local preferences. The bulk material is easy to handle and can be used on land or transported offshore as needed. Standard bulking and cementing equipment were used in the field.

The geopolymer system has been successfully introduced for conductor and surface casing operations, as well as for remedial jobs such as top jobs. Job evaluation procedures, including post-job pressure matching and cement bond evaluation, are similar to those for regular cement jobs.

Using case studies from the UAE, this paper presents the introduction of the low-carbon geopolymer system as an innovative and sustainable alternative to Portland cement. The planning, execution, and evaluation of these cement-free jobs demonstrate its application as a suitable alternative to Portland cement in oilfield operations.

Introduction

Portland cement has been the primary material for cementing oil and gas wells for many years. Its widespread use is attributed to its availability, cost-effectiveness, and well-established performance. However, from a sustainability perspective, Portland cement presents significant environmental challenges. The production of Portland cement is a major source of CO_2 emissions, contributing to global warming and climate change. This process involves the calcination of limestone (calcium carbonate), which releases a substantial amount of CO_2 into the atmosphere. These emissions not only impact air quality but also contribute to the greenhouse effect, exacerbating environmental degradation.

In light of these concerns, geopolymers have emerged as a promising alternative to traditional Portland cement. Geopolymers are aluminosilicate materials that have demonstrated proven results in various applications, including oil and gas well cementing. Unlike Portland cement, geopolymers have a significantly lower carbon footprint, making them a more sustainable option. The hardening mechanism of geopolymers involves the reaction of aluminosilicate with activators in an alkaline medium. This reaction forms a three-dimensional network of aluminosilicate gels, which provides the necessary strength and durability for well cementing applications.

The adoption of geopolymers in the oil and gas industry not only addresses the environmental concerns associated with Portland cement but also offers a viable and effective solution for well cementing. As the industry continues to seek sustainable alternatives, geopolymers present a compelling case for reducing the carbon footprint and mitigating the environmental impact of well cementing operations.

This project was initiated with comprehensive testing conducted at both district and central research facilities using locally sourced raw materials. This rigorous approach ensured that all variables were meticulously evaluated, creating a solid foundation for the subsequent phases of the project. The initial focus was on targeting less challenging, basic jobs with slurry designs, which allowed the refinement of methodologies and the optimization of resources. To further validate the designs, two yard tests were conducted to evaluate different geopolymer-based designs. These tests were performed both on the fly and through batch mixing, providing valuable insights into the performance and adaptability of the materials under various conditions.

Following the successful yard tests, three jobs were carried out, consisting of two conductor jobs and one surface casing job. Each of these jobs was carefully planned and executed, significantly contributing to the successful implementation of replacing conventional Portland cement with geopolymer-based systems.

The details of these tests and job executions, along with the insights gained, will be discussed in greater detail in this paper.

Low-Carbon Geopolymer

Geopolymers are inorganic, environmentally friendly materials synthesized through the reaction of aluminosilicate materials with an alkaline solution. The production process involves several steps, typically carried out at or near room temperature, making it significantly more energy-efficient than traditional cement production.

The primary structural difference between geopolymers and cement lies in the nature of their chemical bonds. Portland cement is based on hydration reactions and ionic bonding, whereas geopolymers consist of three-dimensional covalent bonds formed through the polycondensation of inorganic aluminosilicate solutions (Mark Meade et al. 2023).

As described by Nelson and Guillot (2006), Portland cement sets and hardens primarily through hydration reactions. When mixed with water, Portland cement undergoes a chemical reaction called hydration, in which the cement's compounds react with water to form hydration products. These hydration products, mainly calcium silicate hydrate (C-S-H), gradually fill the spaces between the cement particles, forming a dense, interconnected matrix. This process is responsible for the strength and durability of Portland cement.

In contrast, geopolymers set and harden through a process called geopolymerization, which involves a chemical reaction between aluminosilicate materials and an alkaline activator. When an aluminosilicate source is mixed with an alkaline solution, the high-pH environment dissolves the aluminosilicate material, releasing silica (SiO2) and alumina (Al2O3) into the solution. These dissolved silica and alumina species undergo condensation reactions, forming an amorphous gel known as aluminosilicate gel. The aluminosilicate gel subsequently undergoes further condensation and structural rearrangement to form a hard, three-dimensional, cross-linked geopolymer network.

Fig. 1 illustrates a simplified depiction of the hardening processes of the two systems (Saurabh Kapoor et al. 2023).

Figure 1—Hardening process of portland cement versus geopolymer system.

Geopolymers offer several environmental benefits, making them a more sustainable alternative to traditional Portland cement. One of their most significant advantages is their lower carbon footprint. Unlike the production of Portland cement, which requires high energy during clinker production (involving high fuel consumption and the calcination process, resulting in substantial CO_2 emissions), geopolymer technology uses a byproduct of an industrial process, hence no or minimum CO2 emissions are produced.

In terms of durability and performance, geopolymers demonstrate excellent resistance to chemical attacks, high temperatures, and other harsh environmental conditions. This resilience is attributed to their three-dimensional amorphous structure and unique mineralogy. Unlike Portland cement, geopolymers contain minimal or no free calcium, making them resistant to carbonation, acid attack, and sulfate degradation. These properties enhance the longevity of structures and reduce the need for frequent repairs and maintenance.

Overall, adopting geopolymers in the construction and oil and gas industries provides a viable solution for reducing the carbon footprint and promoting sustainability. To validate this concept, two yard tests and three field jobs were conducted in the UAE. These initial efforts serve as a foundation for future work, paving the way for gradually replacing Portland cement in oil and gas wells.

Laboratory Testing

Unlike Portland cement, whose behavior is well-documented and understood, geopolymers required extensive sensitivity testing to ensure optimal performance. The laboratory testing involved evaluating the effects of several factors, including temperature, type of activator, density variance, and mixing procedures. Each of these parameters can significantly influence the properties and behavior of the geopolymer, making thorough and precise testing essential. By understanding these sensitivities, the formulations for the jobs were tailored to be fit for purpose.

Before commencing laboratory testing, it was paramount to identify locally sourced aluminosilicates. Various UAE suppliers were contacted to provide samples, which were then screened based on their reactivity and specific gravity (SG). This screening process was followed by rigorous testing conducted in accordance with the API 10B-2 standard. Key screening and sensitivity tests performed are summarized below:

- ***Activator Selection and Impact on Mixing Techniques***
  - Evaluation of solid and liquid activators and their effect on fly/batch mixing methods.
- ***Rheology***
  - Measurement of gel strength development over time.
- ***Blend Addition Timing***
  - Determination of optimal timing for blend addition.
- ***Thickening Time Testing***
  - At expected bottomhole circulating temperature (BHCT).
  - With mix fluid and blend preheated to expected wellsite temperatures.
  - ±10°F BHCT sensitivity.
  - Extended temperature sensitivity based on yard test results.
  - Sensitivity to retarder concentration, activator types, and activator concentrations.
  - Density sensitivity.
  - Evaluation of geopolymer blends contaminated with Portland cement.
- ***Compressive Strength Development***
  - At bottomhole static temperature (BHST) and BHCT.
  - At surface temperature.
  - Using a water bath.
  - Using an Ultrasonic Compressive Strength Analyzer (UCA) and hydraulic press for crush cube testing.

Results Discussion. The type of activator significantly impacted the mixing process in terms of changes in slurry temperature. Solid activators mixed into the blend had only a minor impact on the slurry temperature; however, liquid activators caused a significant rise. This is an important observation that should be incorporated during laboratory slurry testing. Instead of relying on surface temperature, the actual measured temperature must be used in subsequent testing. The thickening time was controlled by adjusting the retarder concentration as well as the type and concentration of the activator.

During conductor slurry design testing, it was observed that the geopolymer system developed gel quite rapidly. This shortcoming was mitigated by adjusting the activator and retarder concentrations. Additionally, it was noted that geopolymer-based systems appeared to exhibit greater stability and better solids suspension compared to conventional cement-based systems (Pernites 2025). Geopolymer cubes weighing 15 lbm/gal were tested for compressive strength after 24 hours at surface temperature. The crushed samples showed a compressive strength of 1,675 psi. Ultrasonic cement analyzer (UCA) tests conducted under downhole pressure and temperature conditions indicated compressive strength values close to 2,000 psi (Tables 1 and 2; Figs. 2 and 3).

Figure 2—Cubes for compressive strength measurement using hydraulic press

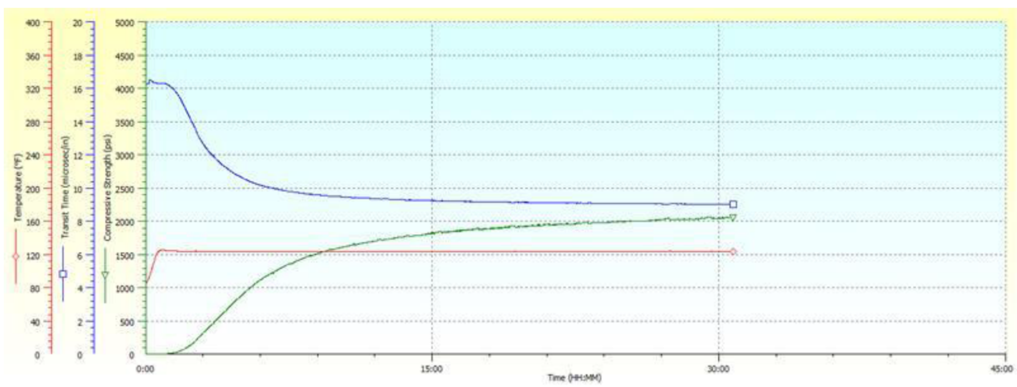

Figure 3—UCA chart for 15.0 lbm/gal geopolymer slurry

Sensitivity testing revealed that the most significant factor influencing thickening time was temperature variation. Increasing the BHCT by 10°F caused a significant reduction in thickening time, whereas decreasing the BHCT by 10°F resulted in a significant increase in thickening time. It is crucial to understand downhole temperature conditions and to simulate BHCT as accurately as possible when using cementing software. Sensitivity to retarder concentration and density showed comparatively smaller variations in thickening time results.

One key observation was the incompatibility between Portland cement and geopolymer systems. Because the same bulk equipment is often used to blend and transport both geopolymer and Portland cement systems, it is critical to thoroughly flush and clean the bulk equipment. However, minor residual contamination in the equipment is unavoidable. To ensure that the slurry remains pumpable, rheology and thickening time tests were conducted with up to 5% contamination of Class G cement in geopolymer blends. The results of these tests are presented in the sections discussing three case histories.

Crush Test Results:

Table 1—Crush test results for 15.0 lbm/gal Geopolymer system for conductor

Time	24:00 hr:mn	
Break Force	**Sample Area**	**Compressive Strength**
6700 lbf	4.00 in2	1,675 psi

UCA Test Results:

Table 2—Ultrasonic Compressive Strength test results for 15.0 lbm/gal Geopolymer system for conductor

Time	Compressive Strength	Temp
24:00 hr:mn	1,974 psi	110 °F

Case Study 1

After capturing the learnings from the yard test with the geopolymer slurry design for the conductor, the first trial job was planned for a 30-in. conductor in a 36-in. open hole. All conductor jobs are performed offline ahead of the rig move. Because of the shallow depth, the slurry in the stab-in conductor job is pumped using centrifugal pumps of the batch mixer instead of a high-pressure pump cement unit. Table 3 shows the job data.

Table 3—Job data for first conductor job with geopolymer

Open Hole Size	36-in.
Conductor Size	30-in.
Section TD	170 ft
Stab-in Shoe	165 ft
Inner String	5 ½-in. DP
Open Hole Annular Excess	75%
BHST/BHCT	110/100 deg F
Slurry Density	15 lbm/gal
Drilling Fluid Density	9 lbm/gal
Fresh Water Spacer Volume	20 bbl
Geopolymer volume	115 bbl

The geopolymer system designed for this job had a density of 15.0 lbm/gal. This slurry system exhibited similar slurry and mechanical properties to the incumbent Portland-based slurry system, which had a density of 15.8 lbm/gal and was previously used for conductor jobs. Laboratory tests conducted for this first job included multiple sensitivity tests, some of which are summarized in Fig. 4. One observation from the yard trial was the increase in slurry temperature when a liquid activator was used in the design. As a result, the slurry was tested at a higher temperature during sensitivity analysis, based on field experience, instead of following the standard practice of ±10°F.

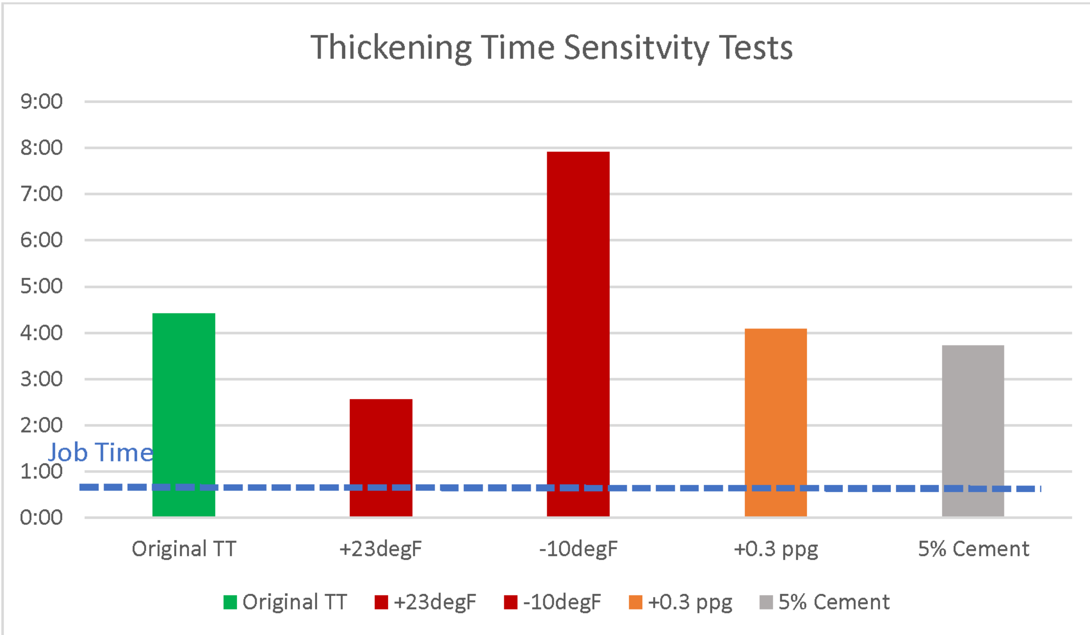

Figure 4—Thickening Time test results with several sensitivity parameters for 15.0 lbm/gal slurry with liquid activator

Geopolymer slurry systems are sensitive to contamination with Portland cement. Therefore, one of the key precautions was to ensure that the bulk equipment was thoroughly flushed and cleaned to minimize contamination. Fig. 5 shows the bulk transport condition before loading the geopolymer blend. Despite the clean bulk equipment, a thickening time test was conducted with 5% Portland cement contamination in a 95% geopolymer system. The results were found to be satisfactory.

Figure 5—Blend transport bulk equipment from inside cleaned before loading

Before preparing the slurry, the temperatures of the mix water and the dry geopolymer blend were recorded. During laboratory evaluation, it was observed that a significant difference between the temperatures of the mix water and the dry blend compared to laboratory conditions could result in misleading results. The temperature of the mix water was found to be 100°F, and that of the dry blend was 105°F, which matched the conditions tested in the laboratory. A total of 120 bbl of slurry was batch mixed in a 200-bbl batch mixer. The density achieved was 15.0 lbm/gal, as per the design, and the volume obtained matched the calculated value.

A total of 120 bbl of the geopolymer system was pumped into the well using centrifugal pumps of the batch mixer. Contaminated slurry returns were observed after pumping 90 bbl, but as pumping continued, neat slurry was obtained at cellar depth, as shown in Fig. 6.

Figure 6—Contaminated geopolymer slurry received on surface through annulus

Surface samples hardened within 6 hours at ambient temperature. Later, when the rig was mobilized to the location, hard cement was drilled on top of the stab-in shoe, similar to conventional Portland cement. The job objective was achieved, and an 85% reduction in overall embodied CO_2 emissions was achieved by switching from Portland cement to a geopolymer-based system (Pernites et al. 2024).

Case Study 2

The second case study also pertains to a conductor job; however, this conductor was for a well designed with a light casing design, where the conductor size was 18 5/8 in. The job execution plan was similar to the first job, with one key difference: the use of solid activators in the design instead of liquid activators. Using solid activators in the blend did not increase the slurry temperature as liquid activators did. Additionally, the slurry mixing process was simpler and faster.

Table 4—Job data for second conductor job with geopolymer

Open Hole Size	26-in.
Conductor Size	18 5/8-in.
Section TD	195 ft
Stab-in Shoe	192 ft
Inner String	5 1/2in. DP
Open Hole Annular Excess	60%
BHST/BHCT	110/100 deg F
Slurry Density	15 lbm/gal
Drilling Fluid Density	9 lbm/gal
Fresh Water Spacer Volume	20 bbl
Geopolymer volume	100 bbl

Sensitivity testing was performed again for temperature, density, and contamination, as shown in Fig. 7. The impact of temperature variation was found to be greater than that of density variation, consistent with the previous job. The geopolymer blend and mix water were heated to the expected field ambient temperature, a practice implemented for all jobs to ensure field conditions were accurately simulated.

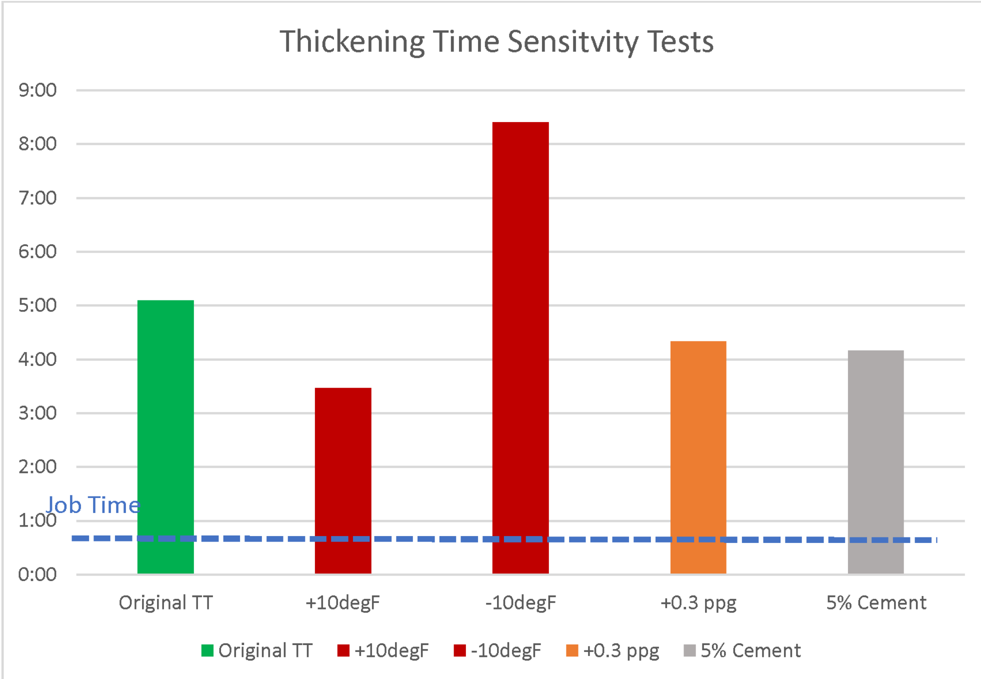

Figure 7—Thickening Time test results with several sensitivity parameters for 15.0 lbm/gal slurry with solid activator

The job was executed by pumping 100 bbl of batch-mixed geopolymer slurry at a density of 15.0 lbm/gal using centrifugal pumps of the batch mixer. Neat slurry was obtained after pumping 85 bbl, after which the remaining 15 bbl was pumped, followed by flushing the surface line with fresh water. Three weeks later, hard cement was drilled when the rig spudded the well, confirming that the job objective was achieved.

Case Study 3

The third job conducted with the geopolymer system was a 13 3/8-in. casing cement job. After the first two jobs were pumped offline ahead of the rig move, this was the first job performed on a rig. The section was drilled with a 17 1/2-in. bit and encountered total losses across a limestone formation near the section's total depth (TD). The planned geopolymer-based system consisted of a single slurry with a density of 15.8 lbm/gal. Job data is provided in Table 5.

Table 5—Job data for surface casing job with geopolymer

Open Hole Size	17 ½-in.
Casing Size	13 3/8-in.
Section TD	1582 ft
Float Shoe/Float Collar	1567/1476 ft
Open Hole Annular Excess	80%
BHST/BHCT	130/115 deg F
Slurry Density	15.8 lbm/gal
Drilling Fluid Density	8.9 lbm/gal
Spacer Density	10.7 lbm/gal
Spacer Volume	40 bbl
Geopolymer Volume	371 bbl
Displacement Volume	221 bbl

During the design stage, several laboratory tests were performed, similar to the previous jobs. Additionally, retarder sensitivity was included in the list of tests. Each bulk truck sample was tested separately to further minimize risk. A summary of the thickening times from these tests is presented in Fig. 8. Compatibility testing with the conventional weighted spacer showed no signs of gelation or mixability issues.

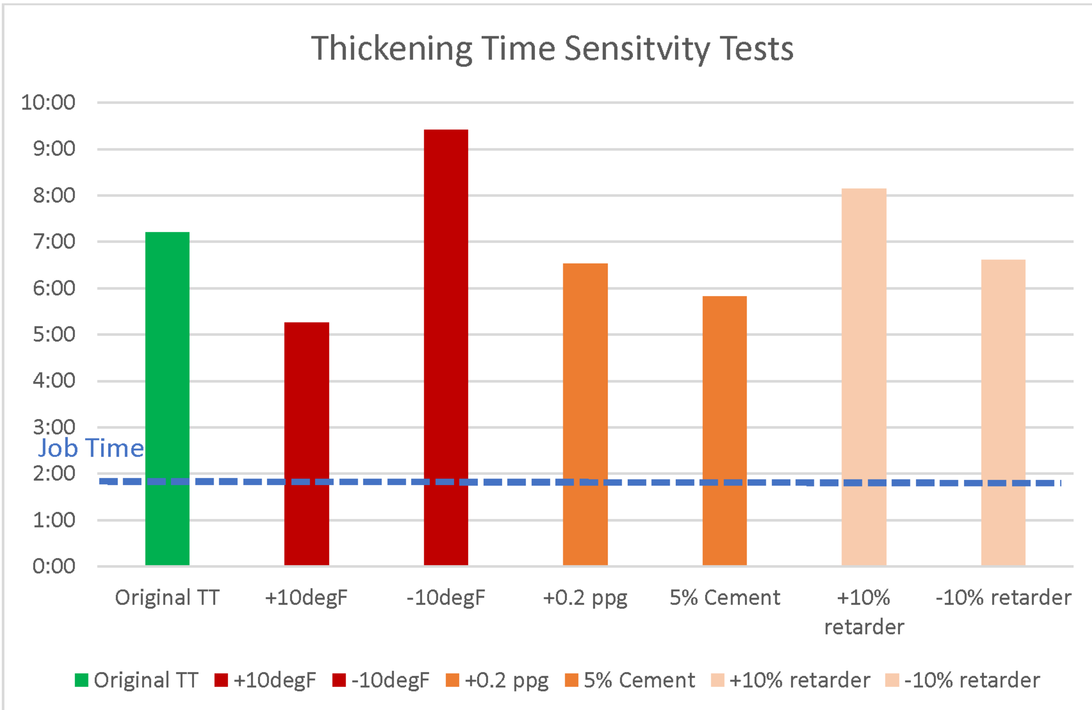

Figure 8—Thickening Time test results with several sensitivity parameters for 15.8 lbm/gal slurry for surface casing

Execution

For the first time, the geopolymer slurry was planned to be mixed and pumped on the fly using standard cementing equipment in the UAE. The mix fluid was prepared in a batch mixer, and the following data was measured and recorded in Table 6:

Table 6—Parameters recorded during job execution of surface casing

Mix Water Temperature	101 deg F
Mix Fluid Temperature	100 deg F
Geopolymer Blend Temperature	109 deg F
Slurry Temperature	120 deg F
Bulk Truck Pressure for blend transfer	Smooth blend flow with 15, 25, 30 psi

The slurry was easily mixable, and the desired density of 15.8 lbm/gal was achieved quickly. Initially, the slurry was mixed and pumped from the smaller 6-bbl tub of the cement pump unit. As the job progressed, the slurry was allowed to overflow into the 14-bbl reserve tub, as shown in Fig. 9. No signs of settling or gelation were observed throughout the job execution.

Figure 9—Mix water temperature being checked, Slurry mixed in cement unit tub, overflowing to reserve tub

Since the well experienced losses while drilling, during geopolymer pumping, returns stopped temporarily after 100 bbl of slurry entered the annulus. However, circulation was regained halfway through displacement. By the end of the job, the spacer was received at the surface. It can be concluded that the geopolymer contributed to curing losses despite the higher equivalent circulating density (ECD) during pumping. Subsequently, a 20-bbl top job was pumped, and neat slurry returns were observed. After 20 hours of waiting on cement (WOC), hard geopolymer was drilled out in the shoe track, and drilling resumed for the next section. The job was successfully completed, achieving the objectives while reducing CO2 emissions by 85%.

Sustainability

Switching to geopolymer-based slurry systems from traditional Portland cement resulted in significant environmental and performance benefits. One of the most compelling advantages is the substantial reduction in embodied CO2 emissions, as geopolymers can reduce CO2 emissions by up to 85% compared to Portland cement. As previously mentioned, this is due to the lower energy requirements and the use of industrial by-products in their production. Using an in-house-developed sustainability assessment tool, the CO2 emissions reduction for the two conductor jobs and one surface casing job was calculated. Droger et al. (2023) discussed using an end-to-end approach to quantify CO2 emissions for the entire cementing operation within this tool, covering the manufacturing of cement, transportation to the rig site, and job execution. The methodology used for this calculation follows the IPCC 2021 Climate Change - Global Warming Potential at 100 years (CC-GWP100) guidelines, as outlined by the Intergovernmental Panel on Climate Change (Salazar et al. 2024). Table 7 and Fig. 10 compare the CO2 emissions produced using Portland cement versus the geopolymer-based system for the two conductor jobs and the surface casing job, along with the overall CO2 emissions reduction achieved across the three jobs.

Table 7—CO_2 emissions comparison between conventional portland based cement and geopolymer system

Scenario	Conductor Job-1, CO_2e (tonnes)	Conductor Job-1, CO_2e (tonnes)	Surface Casing Job, CO_2e (tonnes)	Total CO_2e reduction (tonnes)
Class G Cement	20.7	17.2	64	-88.5
Geopolymer system	2.7	2.3	8.4	0

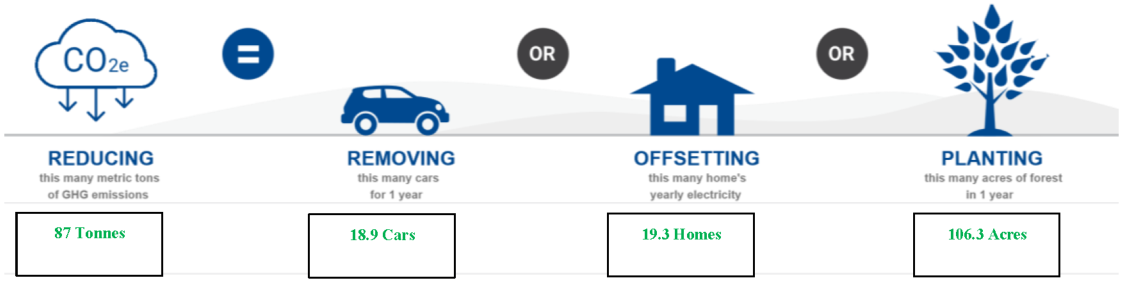

Figure 10—CO2 emissions reduction portrayed with social equivalences

Conclusions

The successful implementation of geopolymer-based systems in three jobs, using standard cementing equipment, has demonstrated their potential as a viable alternative to Portland cement. The planning for these jobs involved screening locally available raw aluminosilicates, followed by extensive laboratory testing that included various sensitivity analyses. While geopolymer systems exhibited slight sensitivity to temperature variations, the impact of density and retarder sensitivity was minimal. The concentrations of retarders and activators played a crucial role in addressing temperature-related sensitivity.

Another important consideration is the potential contamination of geopolymers with Portland cement. Because the same blending and transport equipment is used for job preparation, it is essential to thoroughly clean the bulk equipment before using geopolymer systems.

Data collected from the three jobs indicate that both liquid and solid activators can be effectively used with aluminosilicates. Solid activators resulted in less of an increase in slurry temperatures compared to liquid activators. Both on-the-fly mixing and batch mixing are viable options for preparing and pumping geopolymer systems, similar to conventional cement systems.

The reduction of 87 tonnes of CO_2e from just three jobs highlights the significant environmental advantage of geopolymer systems over conventional cement systems, underscoring their potential to contribute to a greener future.

Acknowledgements

The authors would like to thank the management of ADNOC Onshore and SLB for granting permission to publish this paper. Special thanks are extended to the laboratory teams for their continuous testing efforts, which spanned several months, to ensure that the jobs were executed as per the design. The authors also acknowledge the field execution teams for going the extra mile to perform all necessary checks before and during the execution of the three jobs.

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